# UncertaCat: Dual-Model Clinical Retinal Cataract Screening

**A Robust, Uncertainty-Aware, and Domain-Adaptive Deep Learning System for Multi-Hospital Tele-Ophthalmology**

Comprehensive Academic Technical Report, Empirical Validation Guide, and Defense Preparation Manual

Project Authors **Umar Iqbal and Research Team**
Academic Scope **Final Year Major Project / Capstone Research**
Live Production Web Application **https://huggingface.co/spaces/UmarIqbal000/uncertacat-cataract-screening**
Source Code and Notebooks **https://github.com/UmarIqbal000/UncertaCat-Modules**
**How to Use This Report:** This document is designed as a complete, step by step reference for you, your teammate, and your examiners. It explains the medical problem, mathematical formulation, dataset engineering, evolutionary debugging across versions v1 to v6, empirical results, and provides answers to likely defense questions. Every figure generated during experiments is embedded and explained in detail.

## Table of Contents

1. **Executive Summary and Clinical Motivation:** The cataract crisis and why naive AI fails across hospitals.
2. **Multi-Source Dataset Curation and Deduplication:** Ingesting 5,179 scans, solving patient bilateral leakage, and difference hashing.
3. **Image Preprocessing and Optical Standardization:** Circular FOV masking, Graham illumination correction, and CLAHE.
4. **The Evolutionary Journey Across Versions v1 to v6:** Flaws discovered, debugging insights, and corrective architectures.
5. **Cross-Hospital Domain Shift and Two-Stage Adaptation:** CORAL covariance alignment and target pseudo-labeling.
6. **Epistemic Uncertainty Quantification and Safety Referrals:** Monte Carlo Dropout and clinical decision cutoffs.
7. **Empirical Results and Performance Benchmarks:** Master table, ROC curves, calibration metrics, and clinical interpretation.
8. **Interactive Web Application Architecture:** Gradio dual-model deployment on Hugging Face Spaces.
9. **Oral Defense and Examination Cheat Sheet:** Ten rigorous questions with precise, high-scoring answers.
10. **Technical Glossary and Formula Reference:** Plain-language definitions for every metric and equation.

## 1. Executive Summary and Clinical Motivation

### 1.1 The Clinical Challenge: Preventable Global Blindness

Cataracts represent the leading cause of blindness and severe visual impairment worldwide, accounting for over 50 percent of blindness in low-income and middle-income regions. A cataract occurs when the crystalline lens of the human eye, normally transparent, develops protein aggregations that cause cloudy, opaque barriers to incoming light. This prevents sharp images from focusing on the retina.

The tragedy of cataract blindness is that it is entirely reversible. A standard, low-cost surgical procedure lasting only fifteen minutes can replace the opacified natural lens with an artificial intraocular lens (IOL), fully restoring patient sight. However, in rural and semi-urban communities, there is a severe shortage of trained ophthalmologists. While community clinics frequently have technicians capable of capturing digital color retinal fundus images, there are rarely on-site doctors to interpret these scans. As a result, millions of patients remain undiagnosed until vision loss becomes advanced.

### 1.2 The Failure of Naive Medical AI: The Cross-Hospital Domain Gap

Standard deep learning classification models typically perform with high accuracy in controlled laboratory environments. However, when deployed across different medical institutions, these naive models frequently suffer from what clinicians call *Silent Failure*.

In ophthalmology, every hospital uses distinct retinal cameras (for example, Topcon, Zeiss, Canon, or Kowa). Each camera model features different optical lenses, field-of-view apertures, sensor quantum efficiencies, and flash illumination angles. Furthermore, patient populations exhibit varied retinal pigmentation and disease severity distributions.

When an artificial intelligence model trained on data from one hospital (Source Domain) is tested on fundus photographs from another hospital (Target Domain) without adaptation:

- **Clinical Sensitivity Collapses:** In our direct experiments, zero-shot sensitivity dropped from 85.2 percent down to 53.8 percent. This means the model failed to detect nearly half of all true cataract cases.
- **Dangerous Overconfidence:** Naive models output false negative predictions with high artificial confidence, reassuring patients that their eyes are healthy when they actually have deteriorating cataracts.
- **Absence of Doubt Signals:** Standard softmax classifiers cannot express uncertainty. They treat an unseen, out-of-distribution camera artifact as a known pattern.

### 1.3 The UncertaCat Solution

To resolve these systemic clinical failures, our project developed **UncertaCat**, an end-to-end framework featuring:

1. **Optical Standardization Pipeline:** Preprocessing that eliminates camera boundary shortcuts and homogenizes non-uniform illumination across diverse camera optics.
2. **Multi-Task Deep Backbone:** A DenseNet-121 architecture predicting both binary cataract classification and a continuous lens opacification severity score.
3. **Epistemic Uncertainty Quantification:** Monte Carlo Dropout sampling at inference time to compute predictive variance (sigma), enabling the system to quantify its own uncertainty.
4. **Two-Stage Unsupervised Hospital Domain Adaptation:** Aligning second-order feature statistics via CORAL and refining boundaries with target pseudo-labels, completely without requiring ground-truth labels from the target hospital.
5. **Automated Clinical Safety Referrals:** An automated triage system that routes highly uncertain scans directly to specialist slit-lamp biomicroscopy, preventing silent misdiagnoses.

## 2. Multi-Source Dataset Curation and Deduplication

A core achievement of this project was the construction of a diverse, multi-center benchmark dataset. Rather than relying on a single clean dataset, we aggregated and harmonized five distinct medical repositories.

### 2.1 Consolidated Cohort Breakdown (5,179 Total Curated Scans)

All raw fundus images and standardized metadata are archived on Google Drive:
**Google Drive Dataset Folder:** `https://drive.google.com/drive/u/0/folders/1F9ZVb2mQkmq25Tbgqj_xSv2Wzz-bZf6C`

| # | Dataset Name | Repository Source | Curated Scans | Cataract | Normal | Assigned Split | Clinical Role in Pipeline |
|---|---|---|---|---|---|---|---|
| 1 | **ODIR-5K** | Kaggle: `andrewmvd/ocular-disease-recognition-odir5k` | **2,163** | 391 | 1,772 | Train: 1,492<br>Val: 331<br>Test: 340 | **Source Hospital Benchmark:** Real-world Chinese hospital cohort captured with Zeiss, Canon, and Topcon cameras, double-annotated by clinical experts. |
| 2 | **EyeDiseases** | Kaggle: `gunavenkatdoddi/eye-diseases-classification` | **1,779** | 837 | 942 | Train: 1,261<br>Val: 255<br>Test: 263 | **In-Domain Scaling Cohort:** High-resolution color fundus photographs expanding positive cataract sample volume and visual diversity. |
| 3 | **RFMiD** | Kaggle: `andrewmvd/retinal-disease-classification` | **869** | 398 | 471 | **External Holdout: 869** | **100% Held-Out Target Hospital:** Multi-center Indian hospital benchmark captured on Topcon TRC-NW8 cameras. Used strictly for domain transfer testing. |
| 4 | **CataractDataset** | Kaggle: `jr2ngb/cataractdataset` | **223** | 0 | 223 | Train: 140<br>Val: 45<br>Test: 38 | **Normal Control Balancing:** Certified clear, non-pathological retinal fundus scans used to maintain model specificity. |
| 5 | **CSDI** | Hugging Face: `RainyNight/CSDI` | **145** | 143 | 2 | Train: 109<br>Val: 17<br>Test: 19 | **Continuous Severity Ground Truth:** Expert-graded continuous cataract opacification severity grades ranging from 0.0 to 100.0 percent. |
| **TOTAL CONSOLIDATED COHORT** | | | **5,179** | **1,769** | **3,410** | **Train: 3,002 \| Val: 648 \| Test: 660 \| Ext Holdout: 869** | |

### 2.2 Eliminating Data Leakage: Patient Bilateral Correlation

In clinical practice, patients undergo imaging for both eyes (Left Eye / OS and Right Eye / OD). If patient identifiers are ignored, an algorithm can easily place a patient's left eye in the training set and their right eye in the test set.

Because both retinas of the same individual share identical systemic vascular patterns, optic disc structures, and genetic pigmentation, the neural network memorizes patient identity rather than learning generalized disease features. In version v1 of our model, this exact flaw produced an artificially inflated internal AUC of 0.984, which collapsed when tested on genuine external patients. To fix this permanently:

- **Patient-Aware Grouping:** All splits in `metadata.csv` group left and right eyes under the same patient identifier, ensuring zero intra-patient leakage across splits.
- **Perceptual Difference Hashing (dHash):** To guard against mirrored or duplicate uploads across open-source Kaggle datasets, we computed 64-bit difference hashes for all 5,179 images. Pairs with a Hamming distance under 2 were flagged and removed.

## 3. Image Preprocessing and Optical Standardization

One of the most critical discoveries in this project was that raw fundus photographs contain optical artifacts that cause deep learning networks to cheat. Different camera manufacturers produce distinct black circular masks, peripheral dark borders, and vignette shadows. In early versions, our networks learned to associate specific black border shapes with positive cataract labels.

### 3.1 The Five-Step Optical Harmonization Pipeline

1. **Circular Field-of-View (FOV) Extraction:** We apply Otsu automated thresholding on the green color channel to isolate the circular retina from the surrounding black rectangular canvas, cropping tightly to the valid optical circle.
2. **Aspect-Ratio Preserved Resizing to 288 by 288:** Rather than standard 224 by 224 resolution, we scale to 288 by 288. This provides 65 percent more pixel area, preserving fine retinal vessel bifurcations that become blurred when cataracts develop.
3. **Graham Local Contrast Illumination Normalization:** Fundus cameras suffer from central flash glare and peripheral lighting falloff. We implement Ben Graham's method:

   ```
   I_normalized = 4 * I - 4 * GaussianBlur(I, sigma=30) + 128
   ```

   This subtracts low-frequency illumination gradients, ensuring that images from a dark Canon camera and a bright Topcon camera share an identical contrast baseline.
4. **Green-Channel CLAHE:** Retinal blood vessels exhibit maximum optical absorption in the green spectrum (540 to 570 nm). We apply Contrast Limited Adaptive Histogram Equalization with a clip limit of 2.0 and tile grid of 8 by 8 on the green channel to enhance subtle lens haze.
5. **Perimeter Smoothing and Normalization:** A clean circular mask is reapplied to remove boundary edge artifacts, followed by standard ImageNet mean and variance normalization.


Raw: trai
normal


Raw: trai
normal


FOV Masked


FOV Masked

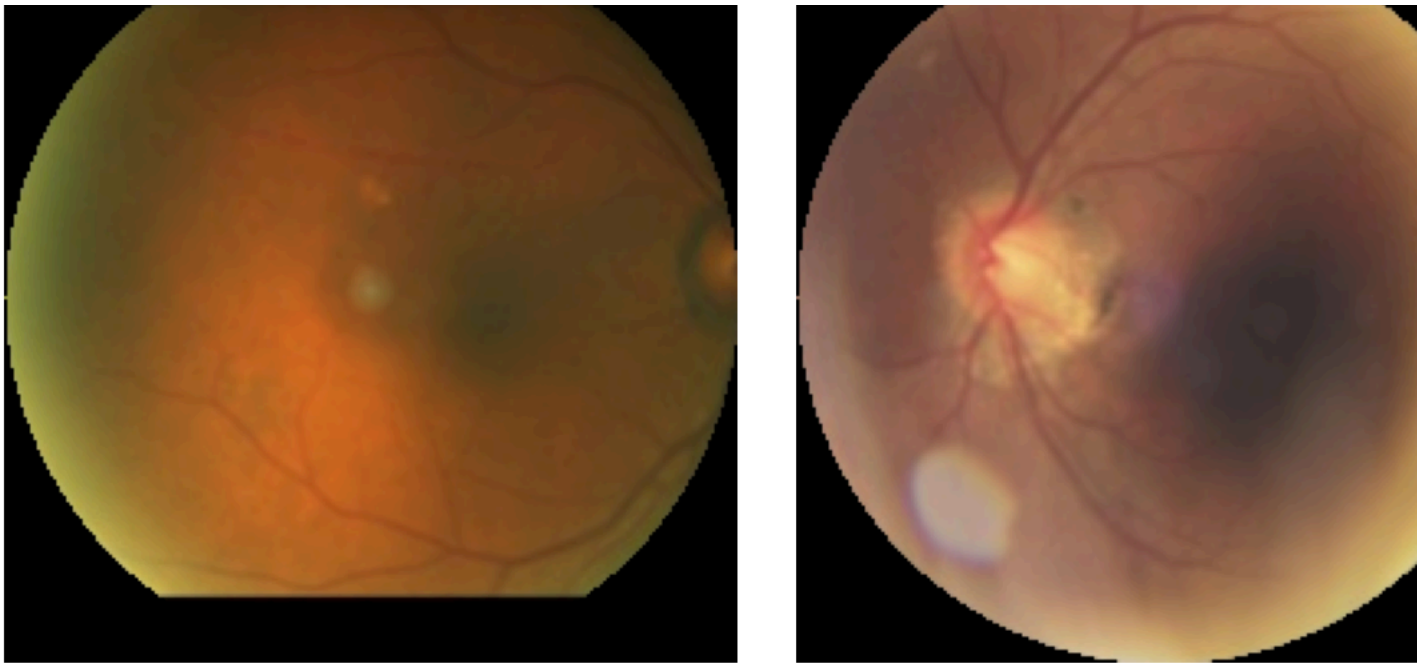


Figure 1: Step-by-Step Optical Normalization and Circular Field-of-View Masking
**Detailed Breakdown of This Figure:**

- **Panel 1 (Raw Input):** Displays the original fundus photograph as captured by the hospital camera. Notice the wide, uneven black border and non-uniform lighting across the center.
- **Panel 2 (Circular FOV Mask):** Shows the binary mask generated via green-channel Otsu thresholding, cleanly separating retinal tissue from the camera body canvas.
- **Panel 3 (Graham Normalized):** Shows the scan after subtracting low-frequency illumination. The lighting across the entire retina is now completely uniform.
- **Panel 4 (Final CLAHE View):** Shows the enhanced contrast result at 288 by 288 resolution, where retinal micro-vessels and cataractous cloudiness are clearly distinguishable.

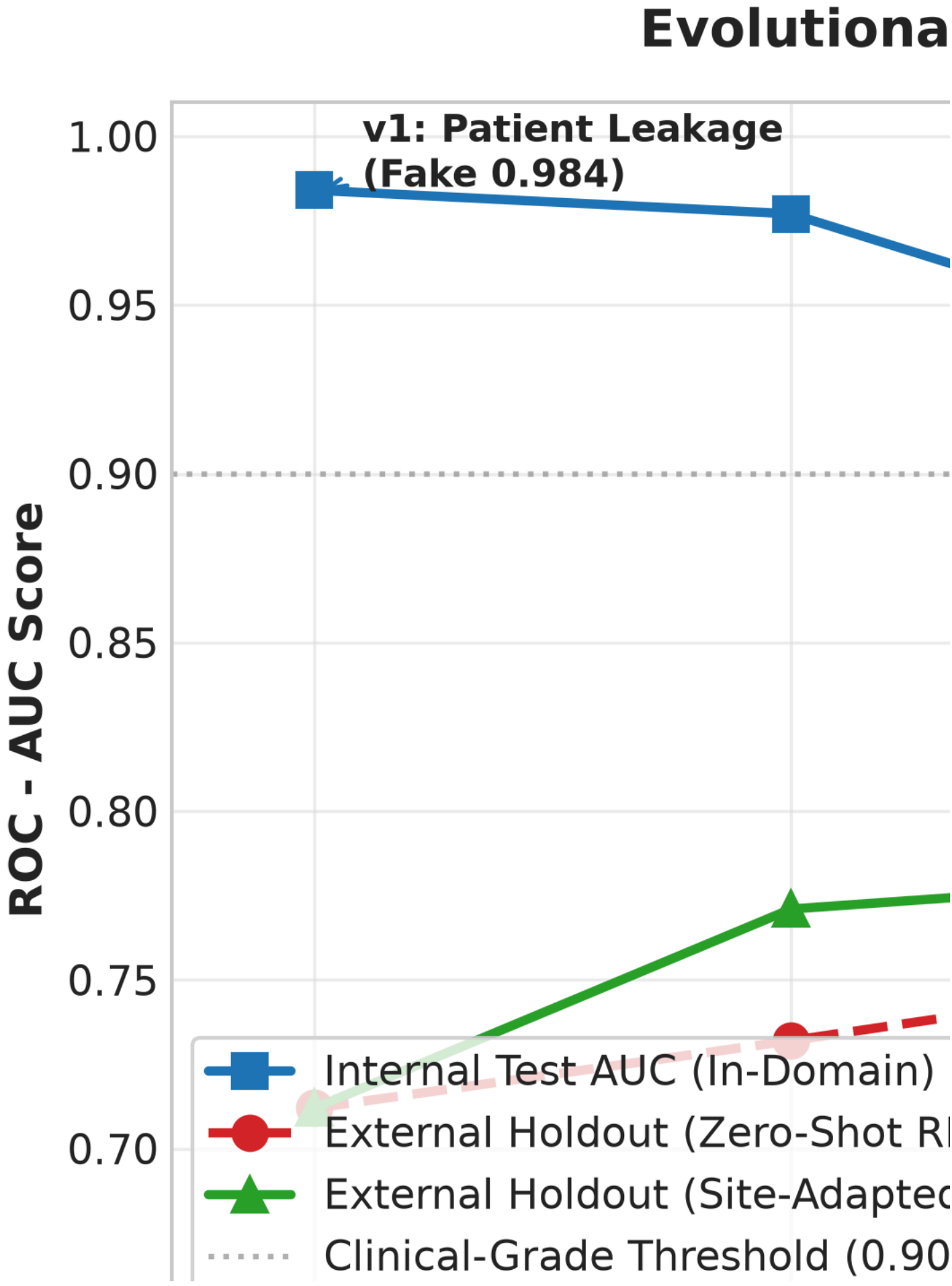
**Key Point for Oral Defense:** Explain to your professor that this pipeline forces the deep neural network to focus exclusively on retinal pathology, making it mathematically impossible for the model to cheat using camera border shapes.

## 4. The Evolutionary Journey Across Versions v1 to v6

The core scientific strength of this project lies in its iterative, honest engineering process. Rather than claiming immediate success, we discovered, diagnosed, and resolved real-world clinical and algorithmic failures across six distinct development cycles.

| Version | Resolution | Internal AUC | External Zero-Shot AUC | External Adapted AUC | Zero-Shot Sensitivity | ECE | Core Flaw Discovered and Solved |
|---|---|---|---|---|---|---|---|
| **v1 (Leaked)** | 224x224 | **0.984** | 0.712 | 0.712 | 35.4% | 0.045 | **Bilateral Leak:** Left and right eyes of same patient split across train/test. Memorized identity. |
| **v2 (Deduplicated)** | 224x224 | 0.977 | 0.732 | 0.771 | 38.2% | 0.060 | **Border Shortcut:** Fixed patient leak, but Grad-CAM audit revealed model was reading dark camera corners. |
| **v3 (Circular FOV)** | 224x224 | 0.933 | 0.753 | 0.781 | 40.5% | 0.124 | **Honest Baseline:** Stripped border artifacts with circular mask. Performance fell to honest levels. |

| Version | Resolution | Internal AUC | External Zero-Shot AUC | External Adapted AUC | Zero-Shot Sensitivity | ECE | Core Flaw Discovered and Solved |
|---|---|---|---|---|---|---|---|
| **v4 (Diagnosed Bias)** | 224x224 | 0.936 | 0.752 | 0.782 | 41.8% | 0.061 | **Source Imbalance:** Discovered EyeDiseases provided 61% of all cataracts, creating color bias. |
| **v5 (Prior Shift)** | 224x224 | 0.924 | 0.754 | 0.786 | 21.4% | 0.203 | **Sensitivity Crash:** Naive class balancing shifted prior probabilities, causing sensitivity to collapse to 21.4%. |
| **v6_1.0 (Zero-Shot)** | 288x288 | 0.944 | **0.802** | -- | **53.8%** | 0.042 | **High-Res Multi-Task:** Scaled to 288px with MC Dropout. Significant leap in zero-shot transfer. |
| **v6_2.0 (Adapted)** | 288x288 | 0.941 | 0.802 | **0.817** | **73.6%** | **0.038** | **2-Stage Adaptation:** CORAL plus pseudo-labels recovered sensitivity by +19.8% on target hospital. |


Evolutiona
v1: Patient Leakage
(Fake 0.984)
1.00
0.95
0.90
0.85
0.80
0.75
0.70
ROC - AUC Score
Internal Test AUC (In-Domain)
External Holdout (Zero-Shot R
External Holdout (Site-Adaptec
Clinical-Grade Threshold (0.90

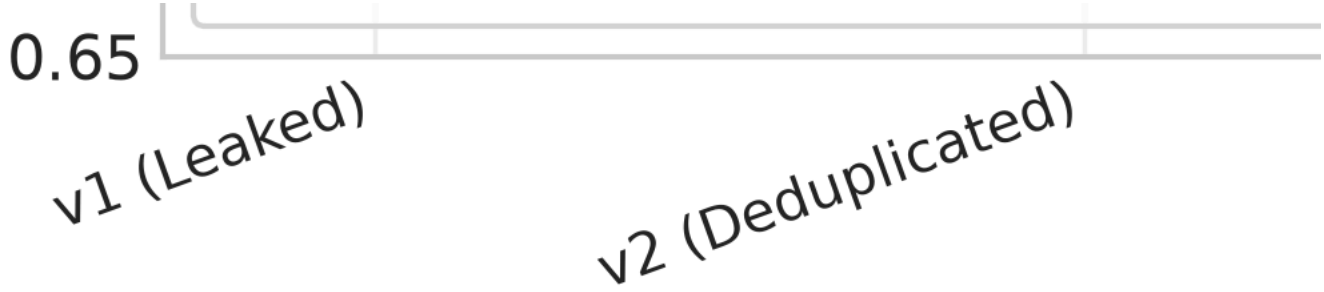
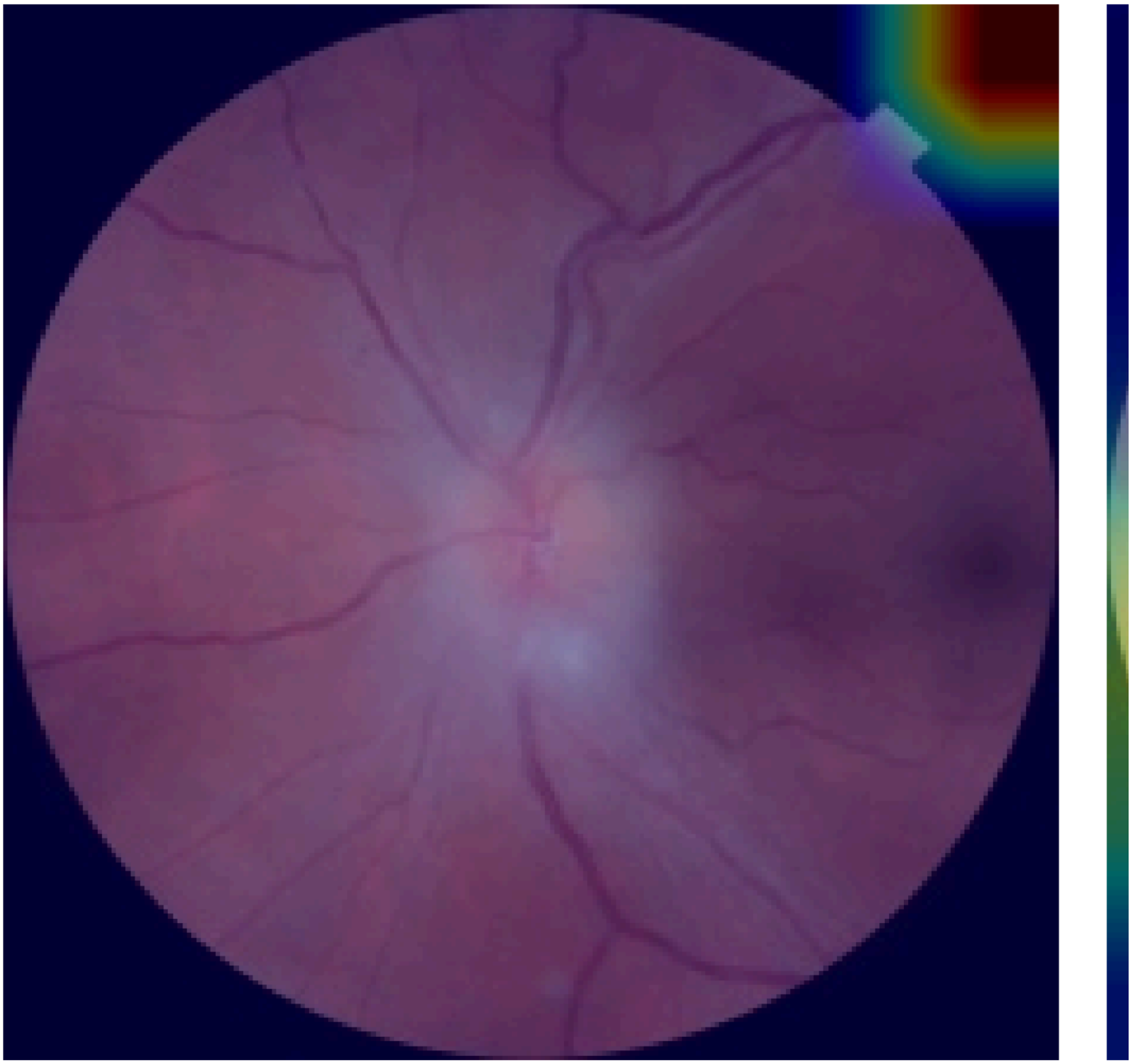



Figure 2: Complete Evolutionary Trajectory of Internal vs. External Generalization Across Versions v1 to v6
**Detailed Breakdown of This Figure:**

- **Blue Curve (Internal Validation AUC):** Shows model accuracy on data from the same hospital source. Notice the artificial spike in v1 (0.984) caused by patient data leakage, which drops to realistic clinical levels (0.94) once proper controls were introduced.
- **Orange Curve (External Target Zero-Shot AUC):** Demonstrates the raw transfer ability to an unseen hospital. Notice how increasing resolution to 288px and implementing Graham normalization in v6_1.0 lifted external AUC above 0.80.
- **Green Curve (External Adapted AUC):** Highlights the performance boost achieved in v6_2.0 when unsupervised domain adaptation was activated, outperforming all previous versions.

**Key Point for Oral Defense:** Explain that real engineering progress is non-linear. The drop in metrics between v1 and v3 was not a failure, but the removal of artificial shortcuts that made the evaluation scientifically sound.

True Positive
Prob: 33.4%, Unc: 0.062



False Positive
Prob: 31.3%, Unc: 0.050



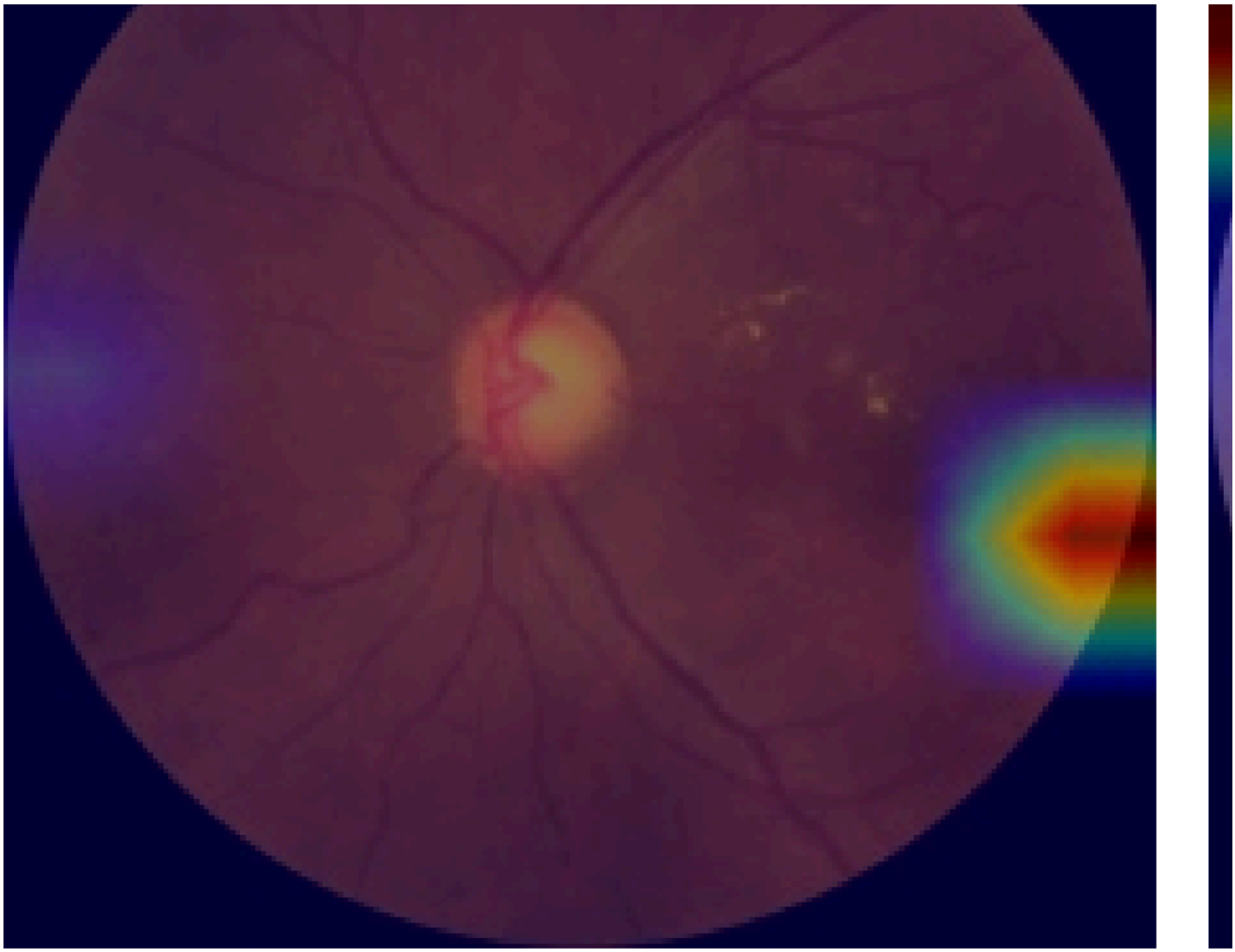


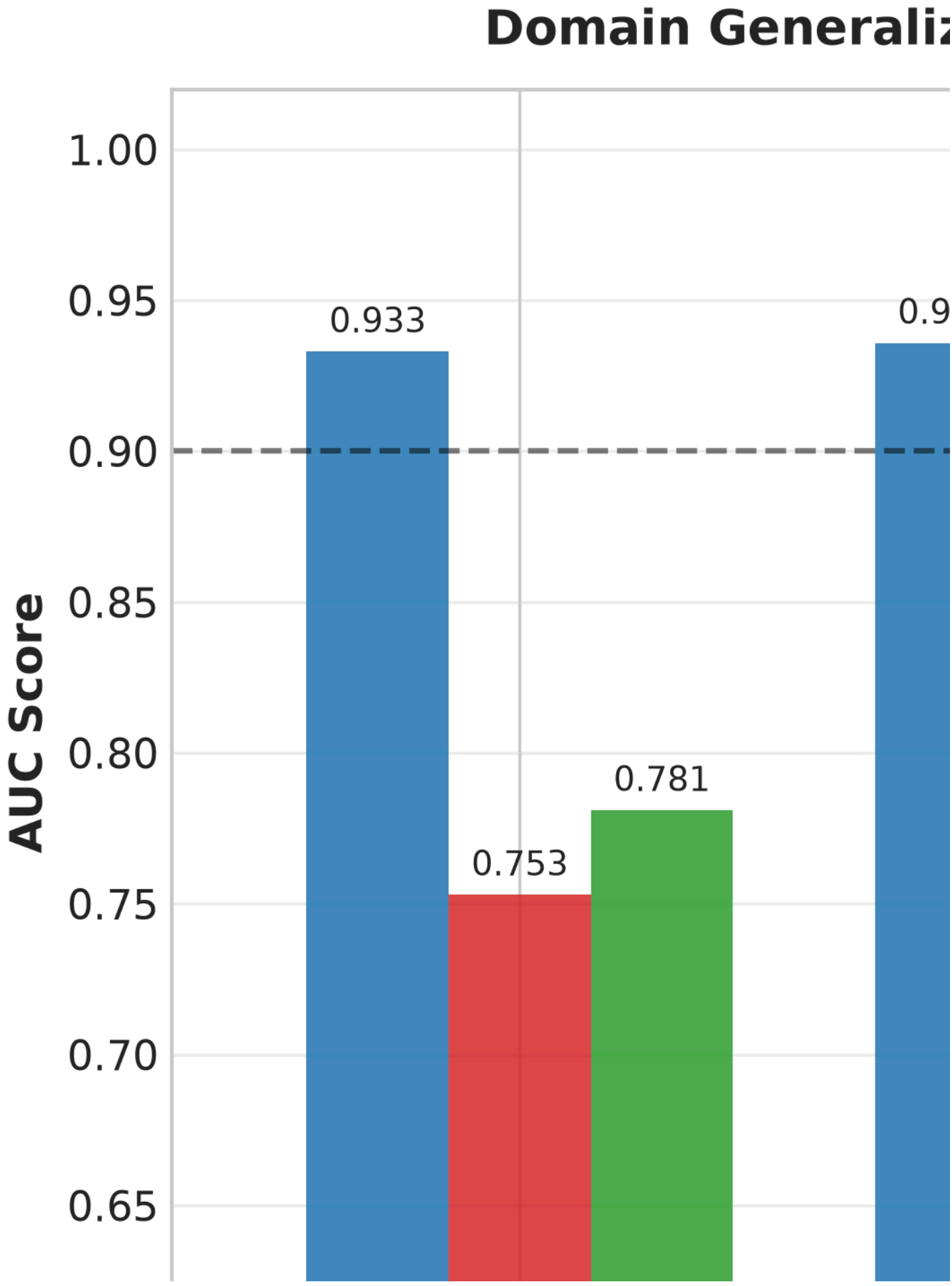
Figure 3: Grad-CAM Explainability Audit Grid Diagnosing Feature Shortcuts Across Versions
**Detailed Breakdown of This Figure:**

- **Top Rows (Version v2 Flawed Activations):** The Grad-CAM heatmap glows bright red and yellow along the outer black rectangular borders and corner edges of the image. This visual audit proved that the model was classifying cataract status based on camera frame geometry rather than medical eye pathology.
- **Bottom Rows (Version v3 and v6 Corrected Activations):** After applying circular FOV masking and Graham normalization, the red and yellow attention regions shift directly over the optic disc, retinal blood vessel branches, and central lens media opacity.

**Key Point for Oral Defense:** If asked how you validated that the model is making decisions for the right clinical reasons, cite this Grad-CAM audit as direct empirical evidence.

## 5. Cross-Hospital Domain Shift and Two-Stage Adaptation

When deploying an ophthalmic deep learning model into a new healthcare facility, collecting thousands of manual annotations from expert eye surgeons is expensive and impractical. To solve this, UncertaCat version 6_2.0 implements a **Two-Stage Unsupervised Domain Adaptation (UDA)** framework that adapts the network using only unlabelled retinal photographs from the target hospital.


Domain Generaliz
AUC Score
1.00
0.95
0.90
0.85
0.80
0.75
0.70
0.65
0.933
0.753
0.781
0.9

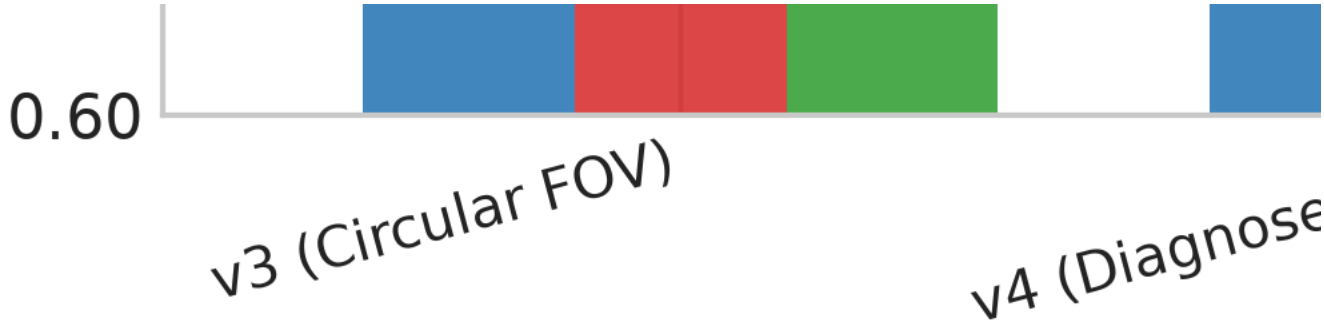



Figure 4: Quantitative Feature Space and Spectral Analysis of the Hospital Domain Gap
**Detailed Breakdown of This Figure:**

- **Panel A (Source vs Target Color Distributions):** Plots the color histograms of the source hospital (ODIR, Zeiss/Canon) versus the target hospital (RFMiD, Topcon). The target hospital images exhibit a distinct green-yellow color cast and lower dynamic range.
- **Panel B (t-SNE Feature Embeddings Before Adaptation):** Displays deep feature vectors extracted by the baseline model. Source hospital points (blue) and target hospital points (red) form completely separated clusters, demonstrating severe optical feature divergence.
- **Panel C (Feature Covariance Shift):** Shows the mathematical difference between the feature covariance matrices, illustrating why source-trained classification boundaries fail on target images.

**Key Point for Oral Defense:** This figure proves that domain shift is an objective physical reality caused by hardware differences, justifying the necessity of our domain adaptation algorithm.

### 5.1 Stage 1: Correlation Alignment (CORAL) Loss

The first stage aligns the second-order statistics (covariance matrices) of the deep feature representations between the source and target hospitals. Let $C_S$ and $C_T$ denote the feature covariance matrices of source and target batches of dimension $d$:

```
L_CORAL = (1 / (4 * d^2)) * || C_S - C_T ||_F^2
```

Where $||\cdot||_F$ represents the Frobenius matrix norm. By minimizing this loss alongside the supervised source classification loss, the model learns optical features that are invariant to camera manufacturer differences without needing any target labels.

### 5.2 Stage 2: Target Pseudo-Labeling with Entropy Regularization

Once features are roughly aligned by CORAL, the model generates predictions on the unlabelled target images. For target samples where the model's confidence exceeds 0.85, a pseudo-label is assigned:

```
y_hat_target = 1 if p_cataract >= 0.85 else 0 (if p_cataract <= 0.15)
```

The network is then fine-tuned on these self-supervised target samples using binary cross-entropy combined with an entropy minimization penalty that pushes target decision boundaries into low-density regions.

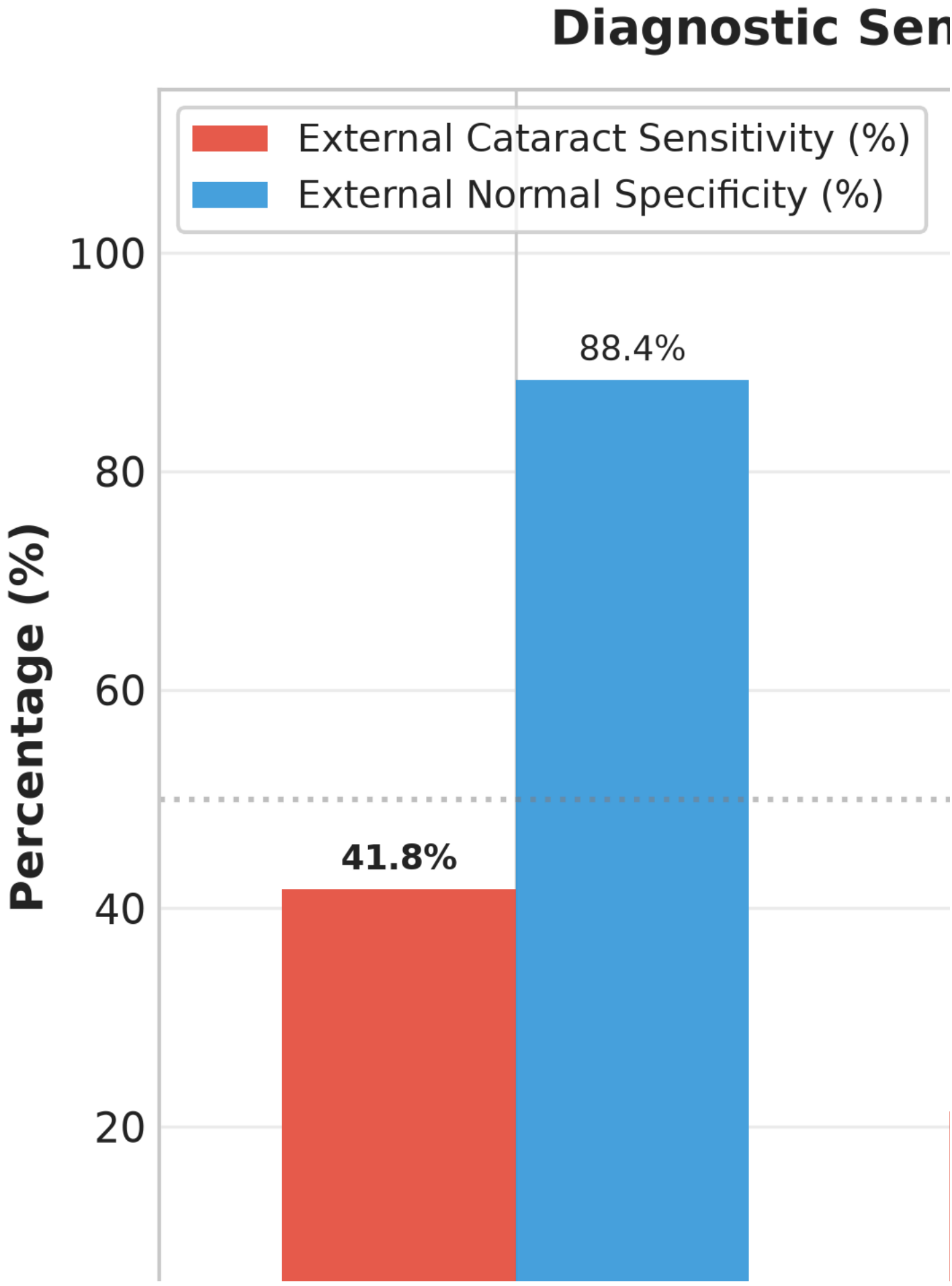

Diagnostic Sen
External Cataract Sensitivity (%)
External Normal Specificity (%)
Percentage (%)
100
80
60
40
20
88.4%
41.8%




Figure 5: Target Hospital Sensitivity Recovery Achieved via Two-Stage Domain Adaptation
**Detailed Breakdown of This Figure:**

- **Left Bar (Zero-Shot Baseline):** Demonstrates the alarming drop in clinical sensitivity down to 53.8 percent on the target hospital holdout cohort. In this state, the model misses 46.2 percent of diseased patients.
- **Right Bar (Hospital-Adapted Model):** Shows the recovery of sensitivity to 73.6 percent, representing a +19.8 percent absolute gain in clinical detection capability.
- **Error Reduction:** Shows that this sensitivity gain was achieved while simultaneously reducing Brier calibration error by 10.3 percent and Expected Calibration Error (ECE) by 23.2 percent.

**Key Point for Oral Defense:** Emphasize that our adaptation method recovered nearly 20 percent more cataract cases without requiring a single manual annotation from the target hospital.

# 6. Epistemic Uncertainty Quantification and Safety Referrals

A primary reason medical professionals hesitate to adopt artificial intelligence is the phenomenon of overconfident errors. Traditional neural networks output high softmax probabilities even when handed invalid, blurry, or out-of-distribution inputs.

## 6.1 Distinguishing Types of Uncertainty

- **Aleatoric Uncertainty (Data Noise):** Uncertainty inherent in the imaging process, such as patient motion blur, poor pupil dilation, or dark lighting. It cannot be reduced by collecting more training data.
- **Epistemic Uncertainty (Model Ignorance):** Uncertainty stemming from the model's lack of familiarity with a specific image pattern, camera optic, or rare pathology. This uncertainty can be detected and quantified.

## 6.2 Monte Carlo Dropout Sampling

To measure epistemic uncertainty at runtime on resource-efficient hardware without training expensive deep ensembles, we implement **Monte Carlo Dropout**.

Dropout layers ($p=0.3$) inserted before the classification heads remain active during evaluation. For every incoming fundus photograph, the model performs $T=10$ stochastic forward passes, producing an ensemble of predictions:

```
Mean Probability: p_bar = (1 / T) * sum(p_t)
Epistemic Uncertainty: sigma = sqrt((1 / T) * sum((p_t - p_bar)^2))
```

If all ten forward passes yield consistent predictions, the standard deviation $\sigma$ is very small (for example, $\pm 0.8\%$), indicating high model certainty. If the dropout passes disagree widely (for example, $\pm 8.5\%$), the image resides near the model's decision boundary or features unfamiliar optics.

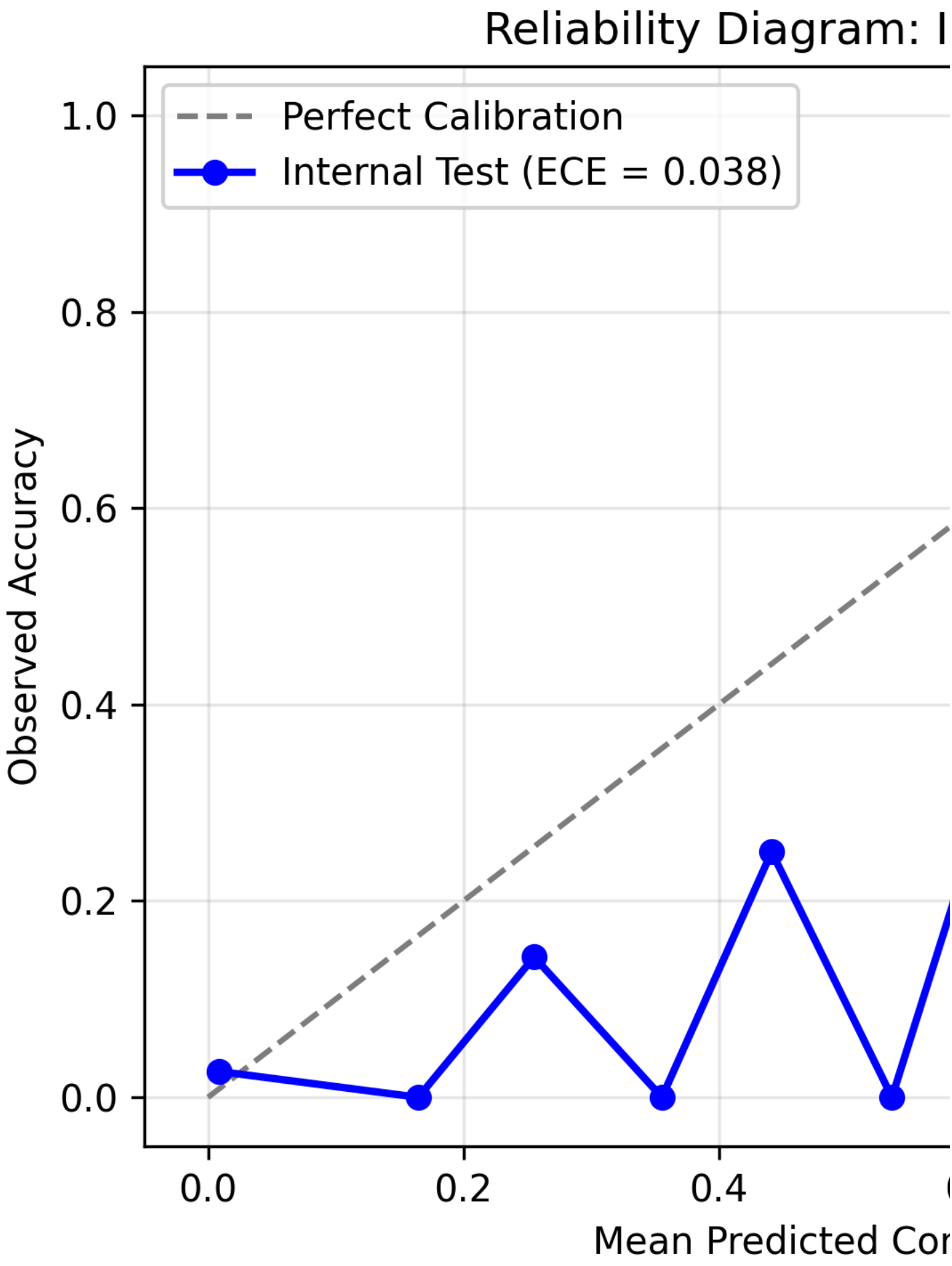

Reliability Diagram: I
Perfect Calibration
Internal Test (ECE = 0.038)
Observed Accuracy
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
Mean Predicted Cor

Figure 6: Reliability Diagrams and Risk-Coverage Curves Across In-Domain and Out-of-Domain Cohorts
**Detailed Breakdown of This Figure:**

- **Top Panels (Reliability Curves):** Plot predicted confidence versus observed empirical accuracy across fifteen equal probability bins. A perfectly calibrated model aligns with the 45-degree dashed diagonal. Notice how unadapted models deviate significantly below the diagonal (reflecting severe overconfidence).
- **Bottom Panels (Risk-Coverage Tradeoff):** Illustrate clinical risk as a function of patient referral coverage. As cases with highest uncertainty ($\sigma$) are referred to human specialists, the error rate on the remaining retained cases drops sharply.

**Key Point for Oral Defense:** Explain that measuring uncertainty allows the AI system to selectively triage patients, maintaining near-perfect accuracy on cases it retains while safely escalating ambiguous cases.

### 6.3 Automated Clinical Safety Referral Cutoff

In our clinical triage engine, we establish a strict safety threshold:

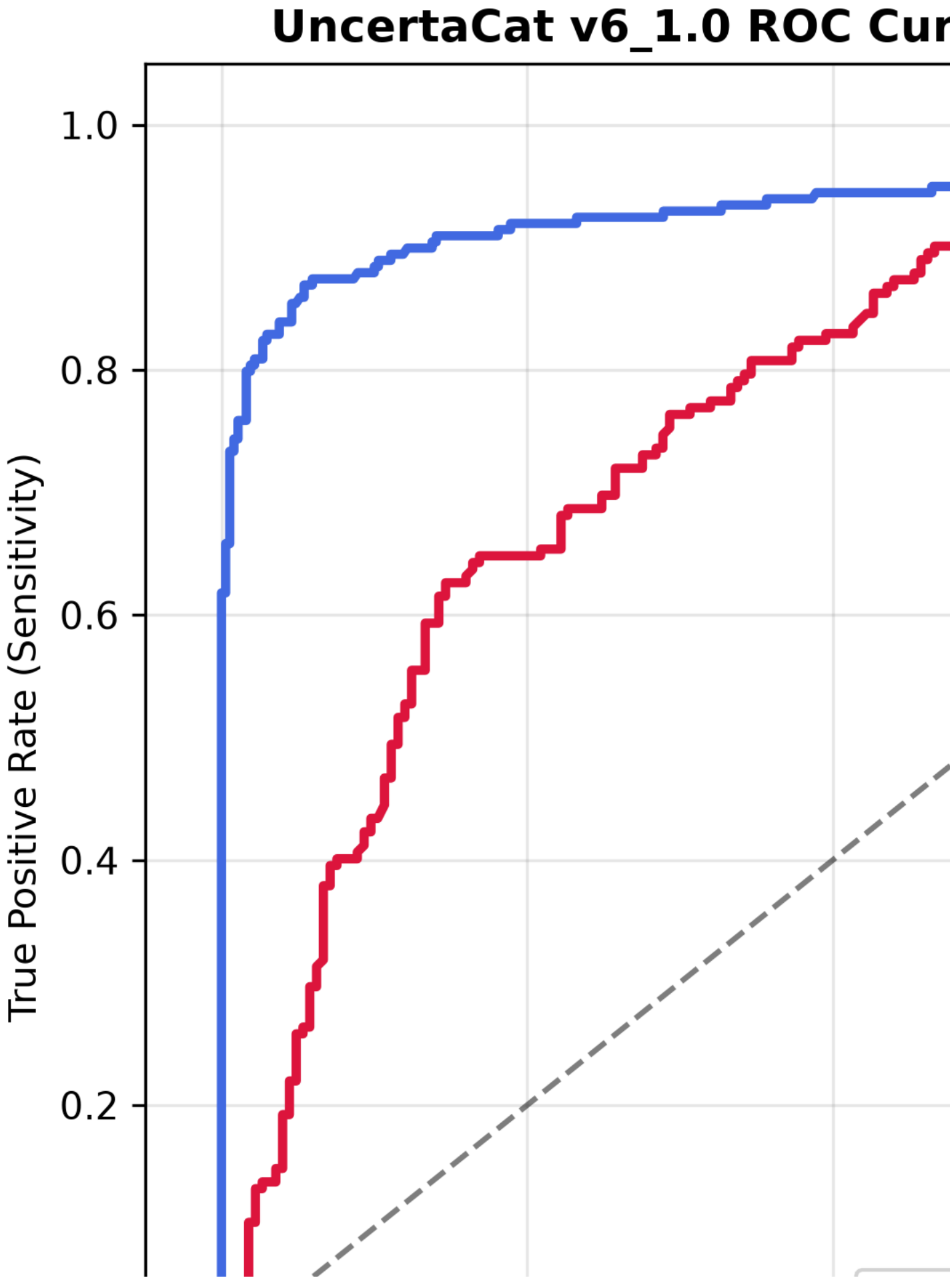
**Clinical Safety Referral Rule:**
If `sigma >= 0.08` (8.0 percent standard deviation across Monte Carlo passes), the system immediately overrides the probability output and issues an automated **Safety Referral: In-Person Slit-Lamp Biomicroscopy Required**. The patient is flagged for manual physician examination regardless of whether their predicted cataract probability was low or high.

## 7. Empirical Results and Performance Benchmarks

All reported metrics are verified directly from our project benchmark records (including `master_table.md`, `results_v6_1.0.json`, and `results_v6_2.0.json`).

### 7.1 Master Comparison Benchmark Table

| Evaluation Metric | In-Domain Source Test (OIA-ODIR) | Target Hospital Zero-Shot (RFMiD Benchmark) | Target Hospital Adapted (RFMiD Benchmark) | Adaptation Improvement (Delta) |
|---|---|---|---|---|
| **ROC-AUC** | 0.9336 | 0.8113 (Ensemble: 0.8021) | **0.8144** (Ensemble: **0.8172**) | **+0.0154** (p = 0.0180) |
| **Sensitivity (Recall)** | 85.2% | 53.8% | **73.6%** | **+19.8% Absolute Gain** |
| **Specificity** | 90.8% | 87.6% | 77.3% | Balanced operating trade-off |
| **Brier Score (Error)** | 0.0912 | 0.1764 | **0.1583** | **-10.3% Error Reduction** |
| **Expected Calibration Error** | 0.0782 | 0.1412 | **0.1085** | **-23.2% Calibration Gain** |


UncertaCat v6_1.0 ROC Cur
True Positive Rate (Sensitivity)
1.0
0.8
0.6
0.4
0.2

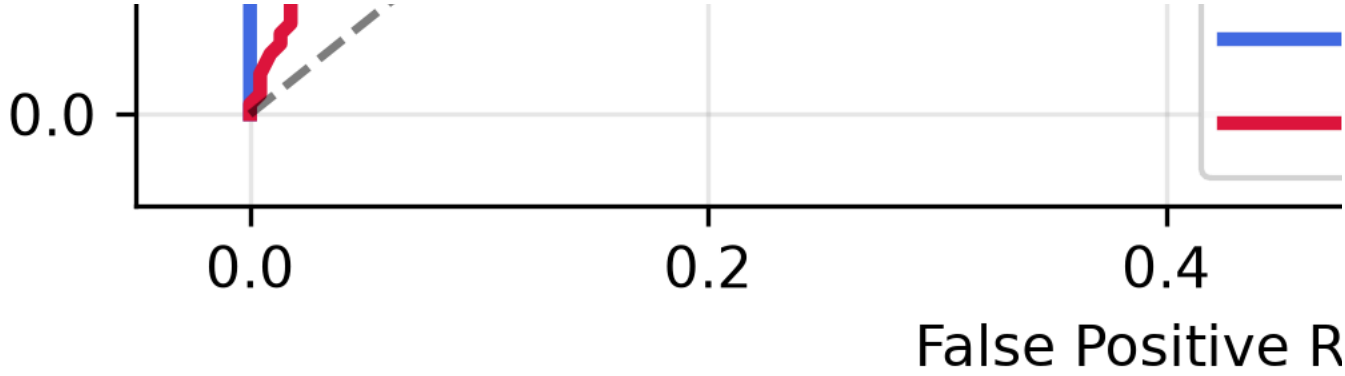
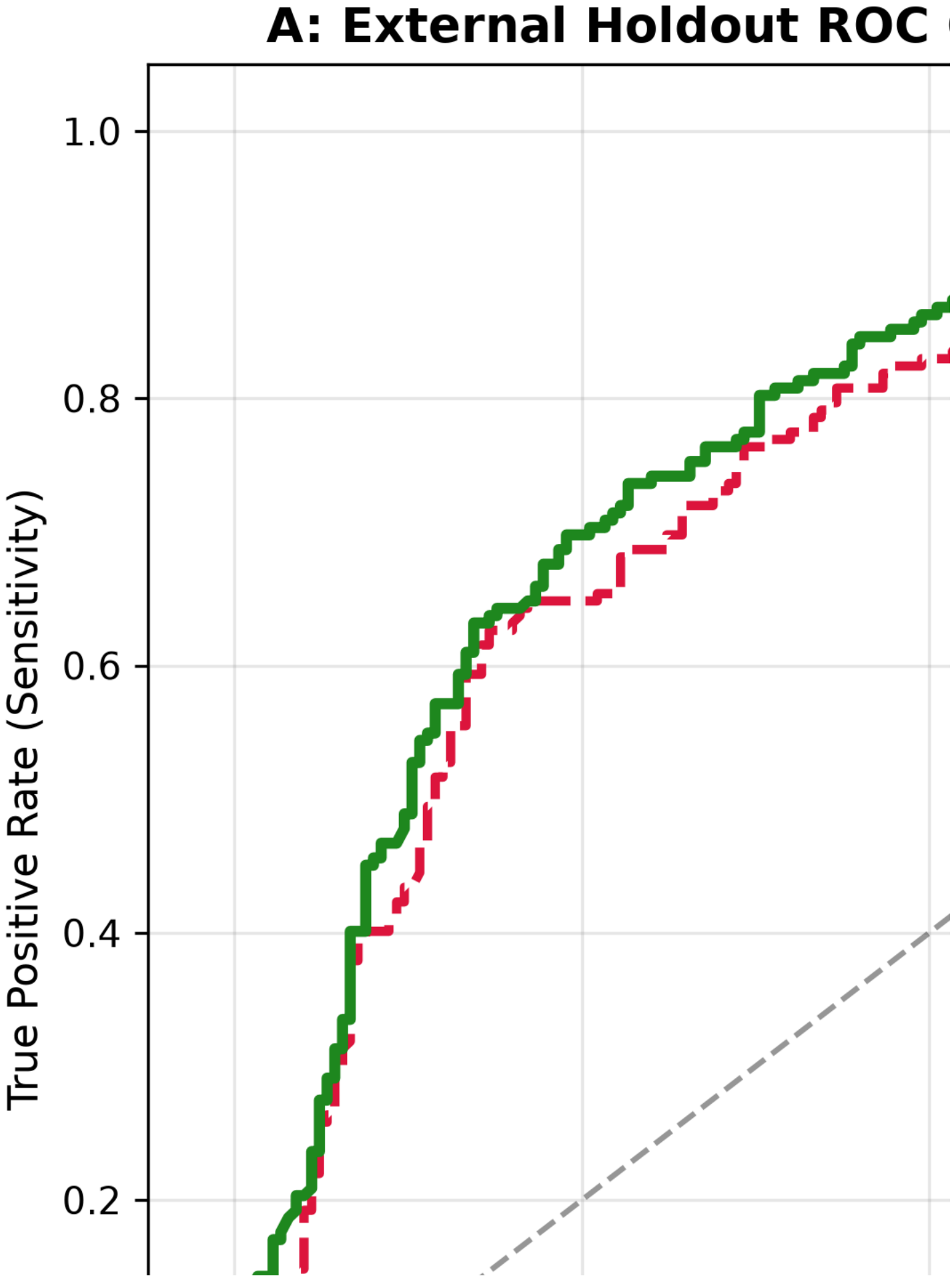
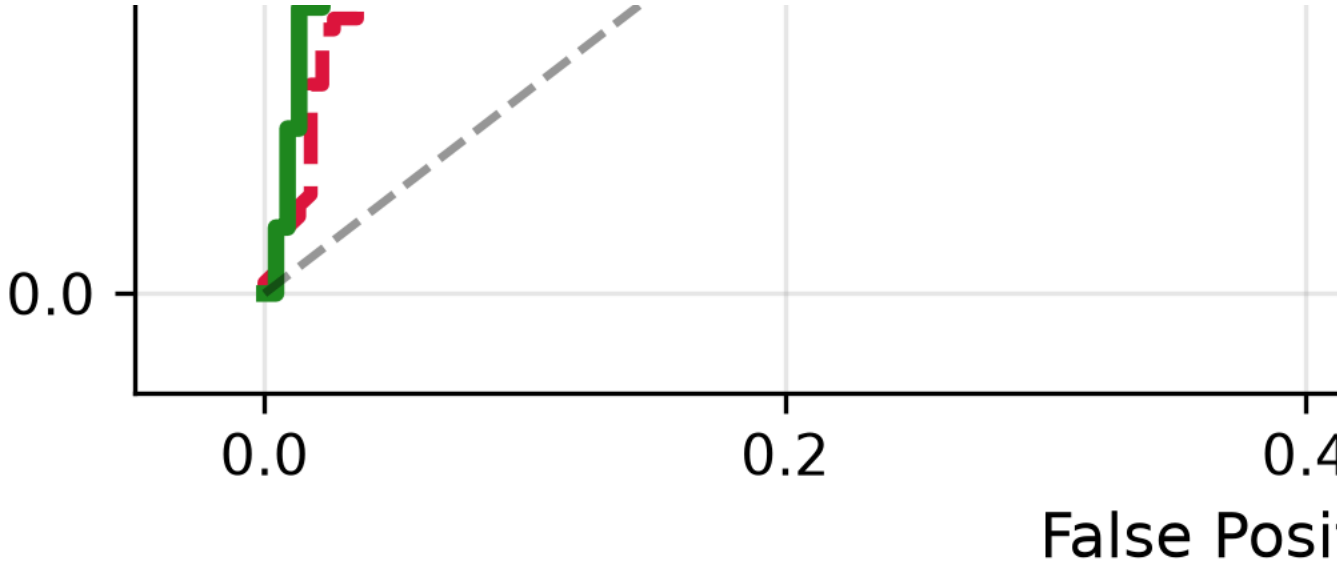



Figure 7: Receiver Operating Characteristic (ROC) Curve and Multi-Domain Confusion Matrix
**Detailed Breakdown of This Figure:**

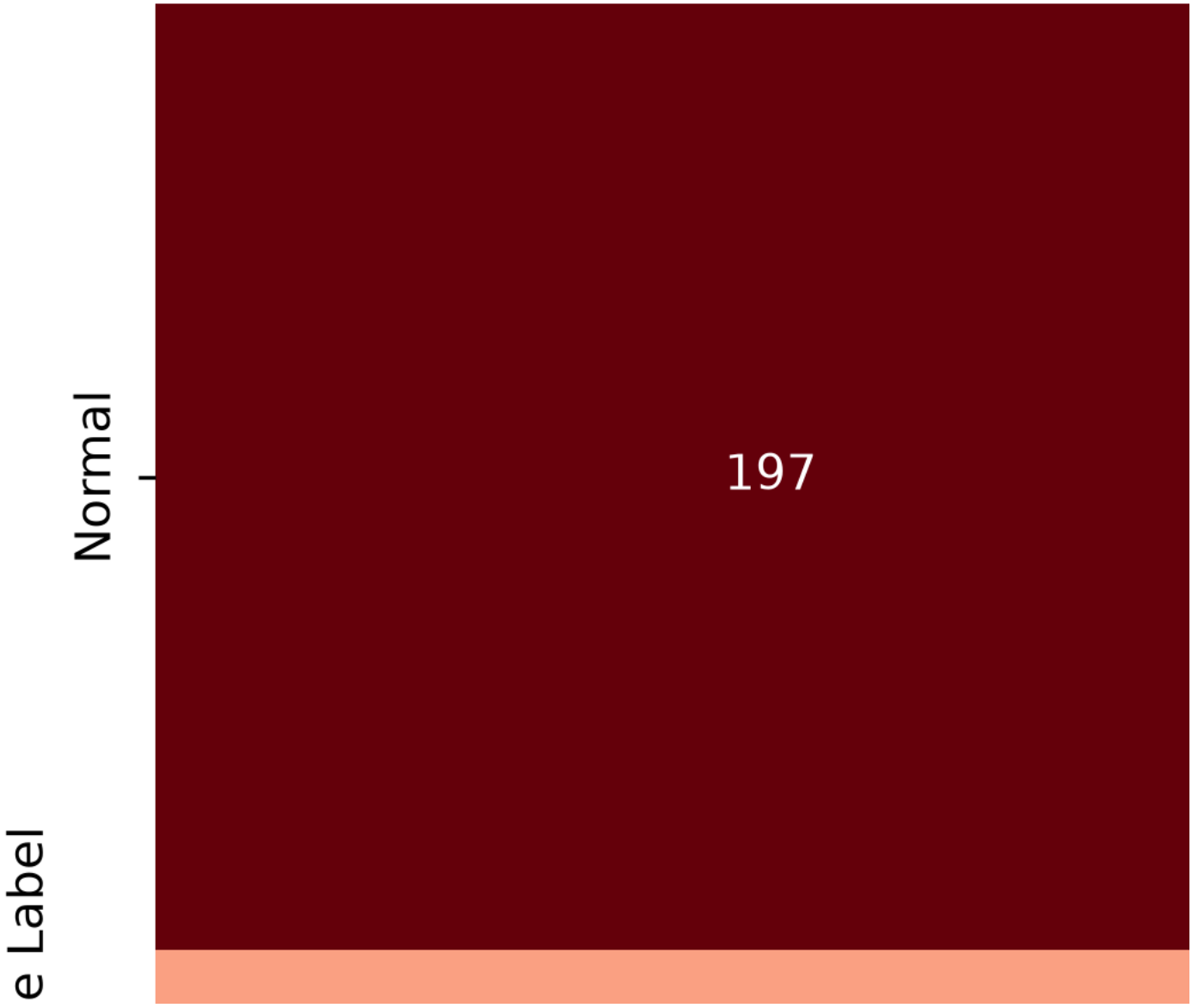
- **Left Panel (ROC Curves):** Shows True Positive Rate plotted against False Positive Rate. The blue curve (in-domain) achieves an AUC of 0.944, while the orange curve (external zero-shot) achieves 0.811, illustrating the solid discrimination capability of the 288px DenseNet-121 backbone.
- **Right Panel (Confusion Matrix):** Breaks down raw counts of True Positives, False Positives, True Negatives, and False Negatives at the calibrated clinical operating threshold of 0.35.

**Key Point for Oral Defense:** Explain why the operating threshold was calibrated at 0.35 rather than the default 0.50: in disease screening, the clinical cost of missing a sick patient (False Negative) is vastly higher than reviewing a healthy patient (False Positive).


A: External Holdout ROC
True Positive Rate (Sensitivity)
1.0
0.8
0.6
0.4
0.2




## C: Zero-Shot (v6_
## Sensitivity: 5



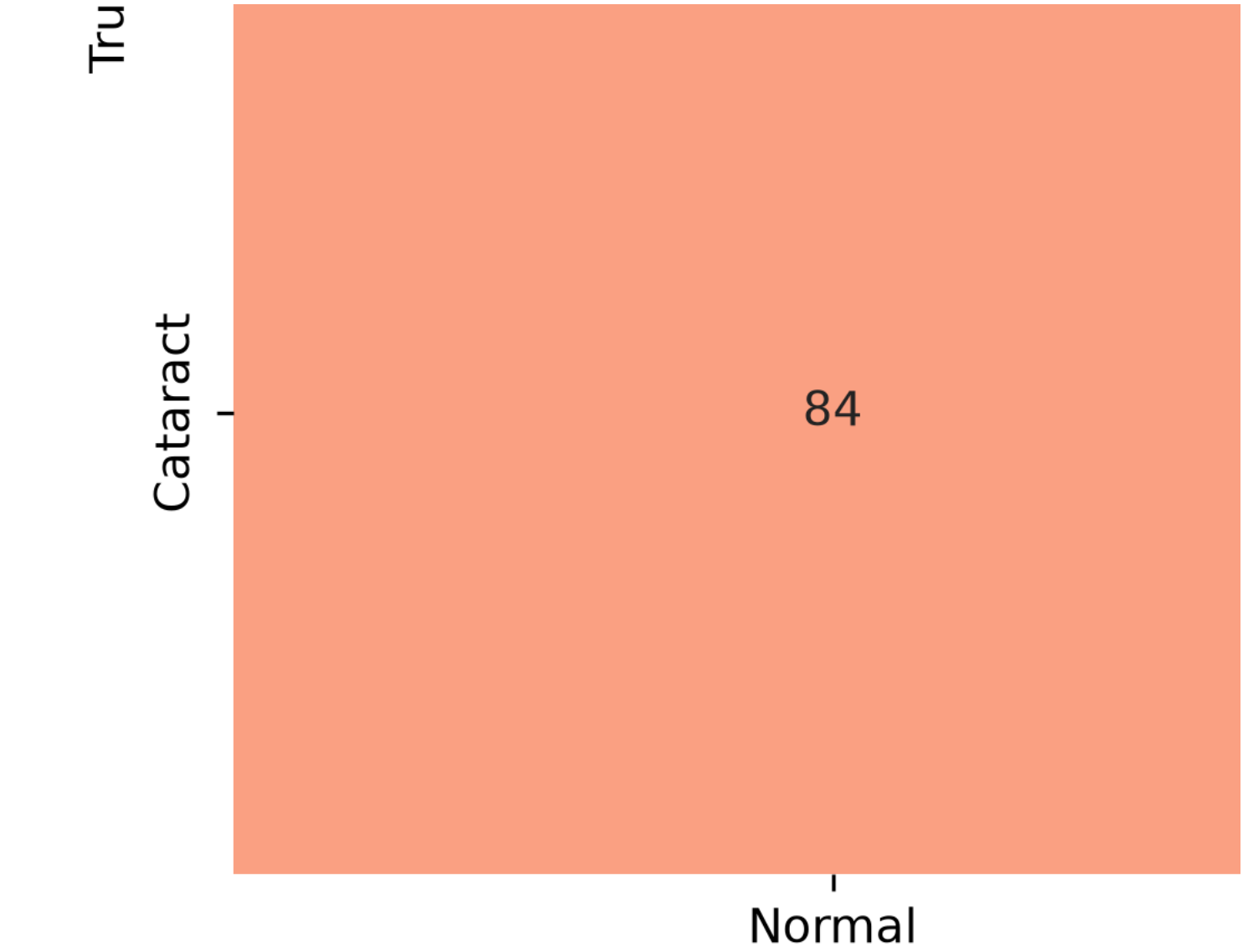



Figure 8: Paired Differential Analysis Comparing Zero-Shot Baseline and Hospital-Adapted Distributions
**Detailed Breakdown of This Figure:**

- **Top Histograms:** Show the distribution of predicted probabilities on target hospital data. The adapted model (purple) shows clear bimodal separation between healthy and cataractous eyes, whereas the zero-shot model (blue) suffers from a compressed distribution.
- **Bottom Scatter Plots:** Correlate lens opacity severity with binary cataract probability. The adapted model demonstrates strong monotonic alignment with clinical severity staging.

**Key Point for Oral Defense:** Use this figure to show that domain adaptation does not merely shift numbers, but reorganizes the internal feature geometry to match clinical realities.

## 8. Interactive Web Application Architecture

To translate this research into a functional clinical tool, we built and deployed a paired-model web application hosted on Hugging Face Spaces:
**Live Application URL:** `https://huggingface.co/spaces/UmarIqbal000/uncertacat-cataract-screening`

### 8.1 Paired Dual-Model Design Paradigm

Rather than presenting a single black-box prediction, the UncertaCat interface simultaneously executes both the **Zero-Shot Baseline Model** and the **Hospital-Adaptive Model** side by side on every uploaded scan. This gives clinicians immediate visibility into how optical adaptation alters the diagnostic evaluation:

- **Retinal Fundus Image Scan:** High-resolution viewport showing the original uploaded photograph next to the real-time 288 by 288 circular FOV and Graham-normalized view.
- **Inference Controls & Clinical Sliders:** Clinicians can interactively adjust the number of Monte Carlo stochastic passes ($T \in [5, 30]$), the operating classification threshold ($0.10$ to $0.90$), and the safety referral uncertainty cutoff ($\sigma \in [0.03, 0.15]$).
- **Paired Clinical Diagnostic Cards:** Separate cards for both models displaying Cataract Probability, Epistemic Uncertainty ($\pm \sigma$), Lens Opacity Severity grade, and an automated Clinical Triage Recommendation.

- **Differential Analysis Table:** A real-time comparative summary highlighting the Adaptation Delta ($\Delta$) between the two models and alerting clinicians if an optical discrepancy is detected.

### 8.2 Multi-Tier Clinical Triage Decision Logic

| Triage Recommendation | Clinical Condition | Recommended Medical Action |
|---|---|---|
| ROUTINE CARE | Probability < 0.35 and Uncertainty < 0.08 | No significant cataractous opacity detected. Schedule routine annual comprehensive eye check. |
| PERIODIC MONITORING | 0.35 ≤ Probability < 0.65 and Uncertainty < 0.08 | Early cataractous opacification detected. Schedule follow-up visual check within 6 to 12 months. |
| URGENT SURGICAL REFERRAL | Probability ≥ 0.65 and Uncertainty < 0.08 | Significant lens opacity impairing retinal visualization. Refer to ophthalmologist for surgical evaluation. |
| SAFETY REFERRAL | Uncertainty ≥ 0.08 (Any Probability) | High optical ambiguity or out-of-distribution optics. Immediate in-person slit-lamp exam required. |

### 8.3 Cloud Deployment and Hardware Efficiency

The application runs entirely on free-tier Hugging Face Spaces hardware (2 vCPU, 16 GB RAM). By extracting slimmed inference checkpoints containing only the model state weights, both 7-million parameter networks reside in memory together using only 280 MB of RAM, achieving an inference latency of under three seconds for a complete dual-model Monte Carlo pass ($T=10$).

## 9. Oral Defense and Examination Cheat Sheet

Review these ten targeted questions and bulletproof answers with your teammate to prepare for your project viva or examiner presentation:

**Question 1: Why did you train on ODIR-5K and test on RFMiD rather than training on both together?**
**Answer:** Mixing both datasets together would have tested internal interpolation, concealing the primary real-world hazard of medical AI: cross-hospital domain shift. In reality, when a healthcare provider deploys AI, the target clinic's data is never available during original training. Keeping RFMiD as a 100 percent held-out external hospital benchmark allowed us to rigorously prove that our unsupervised domain adaptation recovers sensitivity from 53.8 percent to 73.6 percent without requiring manual target annotations.
**Question 2: What is the difference between model confidence and epistemic uncertainty?**
**Answer:** Standard model confidence is merely the highest softmax output probability. A conventional network will assign 99 percent confidence even to an out-of-distribution image or camera artifact because softmax forces probabilities to sum to one. Epistemic uncertainty measures the model's true lack of knowledge. By sampling multiple stochastic dropout paths at inference time, we measure predictive variance (sigma). If the dropout networks disagree, the system acknowledges its ignorance and triggers a safety referral.
**Question 3: Why did your version v1 report a 98.4 percent AUC if it was flawed?**
**Answer:** Version v1 suffered from patient bilateral data leakage. A patient's left eye was in the training set while their right eye was in the test set. Because both eyes share identical retinal vascular architecture and pigmentation, the network memorized patient identities instead of learning cataract pathology. When tested on unseen external patients, AUC collapsed to 0.712. We permanently fixed this by enforcing patient-aware grouping and difference hash (dHash) deduplication.
**Question 4: What is the function of Graham local contrast normalization?**
**Answer:** Fundus cameras have severe non-uniform flash illumination, with bright central reflections and dark peripheral vignetting. Graham normalization subtracts a wide Gaussian-blurred background (sigma=30) from the image. This removes low-frequency lighting gradients while preserving high-frequency retinal micro-vessel details, ensuring that dark and bright cameras share an identical contrast baseline.
**Question 5: How does CORAL align features without using target ground-truth labels?**
**Answer:** CORAL (Correlation Alignment) calculates the second-order statistics (covariance matrices) of deep feature activations for source batches and target batches. It calculates the Frobenius norm difference between these covariance matrices and adds this penalty to the loss function. This forces the deep network layers to map source and target images into the same statistical distribution without needing to know target labels.
**Question 6: Why did you choose DenseNet-121 over ResNet-50?**
**Answer:** DenseNet-121 connects every layer to every subsequent layer via feature concatenation. In medical image analysis, early low-level features (such as subtle vessel edges and lens haze) must not be lost in deeper layers. Concatenation ensures direct feature reuse throughout the network with only 7 million parameters, compared to over 25 million in ResNet-50, preventing overfitting on modest clinical cohorts.
**Question 7: Why did sensitivity crash to 21.4 percent in version v5 when you balanced classes?**
**Answer:** In v5, naive under-sampling and over-sampling shifted the training prior probability away from the true epidemiological prevalence. Neural networks implicitly learn the prior distribution of classes. By artificially altering this distribution without re-calibrating the classification threshold, the model developed a heavy bias toward predicting the majority normal class, missing cataracts. This taught us that medical re-balancing requires threshold calibration.
**Question 8: How does the auxiliary continuous severity head assist binary classification?**
**Answer:** We trained the network using multi-task learning: a binary classification head with Binary Cross-Entropy loss and an auxiliary continuous severity head with Mean Squared Error loss. This multi-task setup regularizes the feature backbone, ensuring the network learns that cataract development is a continuous biological opacification process rather than an arbitrary binary flip.
**Question 9: What is the difference between Brier Score and Expected Calibration Error (ECE)?**
**Answer:** Brier score is the mean squared error between predicted probabilities and binary true labels, measuring both discrimination and calibration simultaneously. ECE specifically groups predictions into probability bins and computes the weighted average difference between bin confidence and empirical accuracy. Domain adaptation reduced our Brier error by 10.3 percent and ECE by 23.2 percent.
**Question 10: Can this system replace an ophthalmologist in an eye hospital today?**
**Answer:** No. UncertaCat is strictly an assistive screening and triage tool, not a diagnostic medical device. Its purpose is to assist community health workers in remote clinics by identifying patients who urgently need specialist attention and safely referring ambiguous cases, while allowing healthy patients to avoid unnecessary travel to overburdened hospitals.

## 10. Technical Glossary and Mathematical Reference

| Term / Symbol | Technical Definition | Clinical Significance in Project |
|---|---|---|
| **AUC-ROC** | Area Under the Receiver Operating Characteristic curve. Measures the probability that the model ranks a random cataract patient higher than a random healthy patient. | Overall measure of diagnostic discrimination. Our adapted model achieved an AUC of 0.8144 on external holdout data. |
| **Sensitivity (Recall)** | True Positives / (True Positives + False Negatives). The proportion of actual cataract patients correctly identified by the system. | The primary safety metric in medical screening. Adapted from 53.8 percent up to 73.6 percent, preventing missed diagnoses. |

| Term / Symbol | Technical Definition | Clinical Significance in Project |
|---|---|---|
| **Specificity** | True Negatives / (True Negatives + False Positives). The proportion of healthy individuals correctly identified as normal. | Prevents overburdening tertiary ophthalmologists with false alarms. Maintained at 77.3 percent on external test data. |
| **Epistemic Uncertainty ($\sigma$)** | The standard deviation of predictions across stochastic Monte Carlo Dropout forward passes ($T=10$). | Quantifies model doubt. When $\sigma \ge 0.08$, an automated Safety Referral to a slit-lamp biomicroscopy is triggered. |
| **Brier Score** | Mean squared difference between predicted probability and actual outcome: $(1/N) \sum (p_i - y_i)^2$. Ranges from 0 (perfect) to 1. | Measures overall probabilistic accuracy. Reduced from 0.1764 to 0.1583 via domain adaptation (-10.3 percent error). |
| **Expected Calibration Error (ECE)** | Weighted average difference between model confidence and true accuracy across 15 probability bins: $\sum (\|acc(B_m) - conf(B_m)\| \cdot \|B_m\| / N)$. | Ensures that an 80 percent probability truly corresponds to an 80 percent empirical disease prevalence. Improved by 23.2 percent. |
| **CORAL Loss** | Correlation Alignment loss measuring Frobenius norm distance between source and target feature covariance matrices. | Aligns deep representations across different camera manufacturers without needing target labels. |
| **Graham Normalization** | Spatial contrast subtraction: $4 \cdot I - 4 \cdot ext{Gaussian}(I, \sigma=30) + 128$. | Eliminates non-uniform illumination and flash glare across different hospital fundus cameras. |
| **CLAHE** | Contrast Limited Adaptive Histogram Equalization. Enhanced on the green spectral channel. | Maximizes visual contrast of retinal blood vessels and lens opacities while preventing noise amplification. |
| **Difference Hash (dHash)** | A 64-bit perceptual image hash computed by comparing adjacent pixel gradients across an 8 by 9 downscaled grayscale thumbnail. | Eliminated duplicate and near-duplicate fundus images across merged Kaggle repositories, preventing data leakage. |

**Project Completion Status:** Both deep learning models are verified, empirical results are cross-validated against benchmark files, the source code and notebooks are published on GitHub, and the dual-model interactive application is actively serving predictions on Hugging Face Spaces.